

Product Requirements Document

Hyper-local Social Media App for India

1. Introduction

This document outlines the detailed product requirements for a hyper-local social media application designed for the Indian market. The app aims to foster strong community bonds, support local economies, and provide a platform for seamless local interaction.

1.1. Product Vision

To be the leading digital platform for every neighborhood in India, enabling residents to connect, share, discover local opportunities, and strengthen their community ties. We envision a platform that empowers local businesses and service providers, making their offerings easily accessible to their immediate community.

2. Target Users

2.1. Primary Users

- **Residents:** Individuals living in a specific neighborhood, seeking to connect with neighbors, discover local events, find local services, and engage in community discussions.
- **Local Businesses:** Small and medium-sized enterprises (SMEs) operating within neighborhoods, looking to promote their products and services, reach local customers, and build a loyal customer base.
- **Local Service Providers:** Individuals or small businesses offering services like plumbing, electrical work, tutoring, home cleaning, etc., within a defined locality.

2.2. Secondary Users

- **Community Leaders/Organizers:** Individuals or groups organizing local events, initiatives, or managing community groups.
- **Local Government/Municipalities:** For disseminating important local information and announcements (potential future integration).

3. User Roles

3.1. Resident

- Can create and manage their profile.
- Can join and participate in neighborhood communities.
- Can post updates, ask questions, and share information within their community.
- Can browse and interact with local business listings and services.
- Can discover and RSVP to local events.
- Can chat with other users (one-on-one and group).
- Can create or join community groups.
- Can receive and manage notifications.

3.2. Business Owner

- Can create and manage a business profile.
- Can list products and services with details and pricing.
- Can post promotions and offers to the local community.
- Can respond to customer inquiries via chat.

- Can manage business reviews and ratings.

3.3. Service Provider

- Similar to Business Owner, but focused on service listings and availability.

4. Features

The application will encompass the following key features:

- **Neighborhood Communities:** Geo-fenced communities allowing users to see and interact with content relevant to their immediate vicinity.
- **Local Business Listings:** A directory of local businesses with profiles, contact information, operating hours, reviews, and service/product offerings.
- **Local Services Marketplace:** A dedicated section for listing and discovering local service providers.
- **Events:** A calendar and listing of local events, workshops, and gatherings. Users can create, promote, and RSVP to events.
- **Chat:** Real-time messaging for one-on-one conversations and group chats within community groups or specific interest circles.
- **Notifications:** Push notifications for new messages, event reminders, community updates, and relevant local alerts.
- **Community Groups:** Users can create or join groups based on shared interests (e.g., gardening, book clubs, sports) within their neighborhood.
- **User Profiles:** Customizable profiles for residents, businesses, and service providers.
- **Content Feed:** A dynamic feed displaying posts, updates, and activities from neighbors, local businesses, and community groups.
- **Search & Discovery:** Robust search functionality to find users, businesses, services, events, and groups.

5. User Stories

5.1. Resident User Stories

- As a resident, I want to find a reliable plumber in my neighborhood so that I can get my leaky faucet fixed quickly.
- As a resident, I want to see what events are happening near me this weekend so that I can plan my leisure time.
- As a resident, I want to connect with my neighbors to discuss local issues or form interest groups.
- As a resident, I want to get recommendations for local restaurants from people living nearby.
- As a resident, I want to receive alerts about local safety or civic updates.

5.2. Business Owner User Stories

- As a local bakery owner, I want to list my shop and showcase my daily specials so that people in my neighborhood know about my offerings.
- As a local store owner, I want to promote a weekend sale to attract customers from my vicinity.
- As a business owner, I want to respond to customer inquiries through the app's chat feature.

5.3. Service Provider User Stories

- As a freelance tutor, I want to list my services and availability so that parents in my area can find me.
- As a home cleaner, I want to manage bookings and communicate with clients through the app.

6. Modules

The application will be structured around the following core modules:

- **User Management:** Registration, login, profile creation and management, authentication.
- **Geo-location Services:** Neighborhood mapping, geo-fencing, location-based content filtering.
- **Community Feed:** Post creation, content display, likes, comments, sharing.
- **Local Directory:** Business and service listings, search, categorization, reviews.
- **Events Management:** Event creation, listing, RSVP, calendar integration.
- **Messaging (Chat):** One-on-one chat, group chat, real-time message delivery.

- **Groups:** Group creation, management, membership, group-specific feeds and chats.
- **Notifications:** Push notification service, user notification preferences.
- **Search & Discovery:** Global search across all modules.
- **Admin Panel:** User management, content moderation, business/service verification, analytics.

7. Functional Requirements

7.1. User Authentication & Profile Management

- FR1.1: Users shall be able to register using mobile number and OTP, or email/password.
- FR1.2: Users shall be able to log in securely.
- FR1.3: Users shall be able to create and edit their profile with name, profile picture, bio, and location.
- FR1.4: Business/Service providers shall have additional fields for business name, address, contact, operating hours, description, logo, etc.
- FR1.5: Implement password reset functionality.

7.2. Neighborhood Communities

- FR2.1: Users shall be automatically assigned to their neighborhood community based on their registered location.
- FR2.2: Users shall be able to view and post content within their assigned community feed.
- FR2.3: Content shall be visible only to users within the same geo-fenced community.

7.3. Local Business & Services

- FR3.1: Verified businesses/service providers shall be able to create detailed listings.
- FR3.2: Users shall be able to search and filter businesses/services by category, name, and location.
- FR3.3: Users shall be able to view business/service details, ratings, and reviews.
- FR3.4: Users shall be able to rate and review businesses/services.

7.4. Events

- FR4.1: Users shall be able to create events with details like title, description, date, time, location, and image.
- FR4.2: Users shall be able to browse upcoming events in their neighborhood.
- FR4.3: Users shall be able to RSVP to events.
- FR4.4: Event creators shall be able to see RSVP lists.

7.5. Chat

- FR5.1: Users shall be able to initiate one-on-one chats with other users.
- FR5.2: Users shall be able to send and receive text messages in real-time.
- FR5.3: Support for basic chat features: typing indicators, read receipts (optional).

7.6. Community Groups

- FR6.1: Users shall be able to create public or private community groups.
- FR6.2: Users shall be able to join/request to join groups.
- FR6.3: Group members shall be able to post content and chat within the group.

7.7. Notifications

- FR7.1: Users shall receive push notifications for new messages, group activity, event reminders, and relevant community alerts.
- FR7.2: Users shall be able to customize their notification preferences.

8. Non-Functional Requirements

8.1. Performance

- NFR1.1: The application shall load within 3 seconds on a stable internet connection.
- NFR1.2: Real-time chat messages shall be delivered within 1 second.
- NFR1.3: The application should handle a minimum of 1000 concurrent users per neighborhood.

8.2. Scalability

- NFR2.1: The architecture shall support scaling to accommodate a growing user base and geographical expansion.
- NFR2.2: The system should be able to handle an increase in data volume gracefully.

8.3. Security

- NFR3.1: All user data, especially personal and sensitive information, shall be encrypted at rest and in transit.
- NFR3.2: Implement robust authentication and authorization mechanisms.
- NFR3.3: Protect against common web and mobile vulnerabilities (e.g., SQL injection, XSS).

8.4. Usability

- NFR4.1: The user interface shall be intuitive and easy to navigate for all user demographics.
- NFR4.2: Support for multiple Indian languages (e.g., Hindi, English initially) will be considered for future phases.

8.5. Reliability

- NFR5.1: The application shall have an uptime of at least 99.5%.
- NFR5.2: Implement proper error handling and logging mechanisms.

8.6. Maintainability

- NFR6.1: The codebase shall be well-documented, modular, and follow established coding standards.

9. Technology Stack

9.1. Mobile Application (iOS & Android)

- **Cross-platform framework:** React Native or Flutter (preferred for faster development and wider reach)
- **State Management:** Redux (for React Native) or Provider/Bloc (for Flutter)
- **Navigation:** React Navigation (for React Native) or GoRouter/Navigator 2.0 (for Flutter)

9.2. Backend

- **Language/Framework:** Node.js with Express.js, Python with Django/Flask, or Golang with Gin (choice depends on team expertise and scalability needs)
- **API Type:** RESTful APIs or GraphQL
- **Real-time Communication:** WebSockets (e.g., Socket.IO)
- **Cloud Platform:** AWS, Google Cloud Platform (GCP), or Azure
- **Containerization:** Docker, Kubernetes (for scalability and deployment)

9.3. Database

- **Primary Database:** PostgreSQL or MySQL (for structured data like user profiles, business listings)
- **NoSQL Database:** MongoDB or Cassandra (for flexible data like chat messages, feed posts - consider if high write throughput is needed)
- **Caching:** Redis (for session management, frequently accessed data)

9.4. Other Tools

- **Push Notifications:** Firebase Cloud Messaging (FCM), Apple Push Notification Service (APNS)
- **Mapping & Geolocation:** Google Maps API, Mapbox API
- **Search:** Elasticsearch (for advanced search capabilities)

10. Database Overview

The database will consist of several key entities:

- **Users:** Stores user profile information, credentials, roles, and location.
- **Businesses/Services:** Stores details of local businesses and service providers, including contact info, categories, hours, and offerings.
- **Posts:** Stores content shared in the community feed, including text, images, and metadata.
- **Events:** Stores event details, dates, times, locations, and RSVP information.
- **Groups:** Stores group information, members, and privacy settings.
- **Messages:** Stores chat messages, sender, receiver, timestamp, and conversation ID.
- **Reviews/Ratings:** Stores user feedback for businesses and services.
- **Notifications:** Stores notification data for users.

Data Relationships: Relationships will be defined using foreign keys (in RDBMS) or embedded documents/references (in NoSQL) to link users to posts, businesses to reviews, users to events they RSVP'd to, etc.

Geo-spatial Indexing: Appropriate indexing will be applied to location data to support efficient geo-queries for neighborhood filtering and discovery.

11. API Overview

The backend will expose a set of RESTful APIs (or GraphQL endpoints) to facilitate communication between the mobile applications and the server.

Key API Endpoints/Operations (Examples):

- (POST: register, GET: get profile, PUT: update profile)
- (POST: login)
- (GET: get feed for current community)
- (GET: list businesses, POST: create business listing, GET: get business details)
- (similar to businesses)
- (GET: list events, POST: create event, PUT: update event)
- (GET: list groups, POST: create group, GET: get group details)
- (POST: send message, GET: get messages for conversation)
- (GET: get user notifications)
- (GET: global search)

Authentication: API requests will be authenticated using JWT (JSON Web Tokens) or session-based authentication.

Versioning: APIs will be versioned (e.g.,) to ensure backward compatibility.

12. Admin Panel Requirements

A web-based admin panel will be developed to manage the platform.

- **User Management:** View, manage, and deactivate user accounts.
- **Content Moderation:** Review and moderate user-generated content (posts, comments, reviews) to ensure community guidelines are followed.
- **Business/Service Verification:** Verify and approve business and service provider listings.
- **Category Management:** Manage categories for businesses, services, and events.
- **Notification Broadcast:** Ability to send broadcast notifications to specific communities or all users.
- **Analytics & Reporting:** View key metrics such as user growth, engagement, popular businesses/events, and system performance.
- **Support Ticket Management:** Handle user support requests.

13. MVP Scope

The Minimum Viable Product (MVP) will focus on core functionality to validate the product concept and gather initial user feedback.

MVP Features:

- User Registration & Profile Management (Resident)
- Neighborhood Community Feed (Posting & Viewing basic text content)
- Basic Local Business Listings (View only, no creation initially)
- Basic Chat (One-on-one text messages)
- Push Notifications (New messages only)
- Geo-location assignment to a neighborhood

MVP Exclusions:

- Business/Service provider profiles and listings creation
- Events management
- Community Groups
- Advanced search and filtering
- Ratings and reviews
- Multiple language support
- Admin Panel (basic functionality only for essential monitoring)

14. Roadmap

14.1. Phase 1: MVP Launch

- Focus on core features as defined in the MVP scope.
- Launch in a pilot city/region.
- Gather user feedback and iterate.

14.2. Phase 2: Feature Expansion

- Implement Local Business/Service Provider Listings with creation and management.
- Introduce Events module.
- Enhance Chat with group chat functionality.
- Develop Community Groups module.
- Expand notification types.

14.3. Phase 3: Monetization & Growth

- Introduce monetization strategies (e.g., featured business listings, local ads).
- Expand to more cities/regions.
- Implement advanced search and discovery features.
- Integrate user reviews and ratings.

14.4. Future Phases

- Local services marketplace enhancements.
- Integration with local government services.
- Advanced analytics for businesses.
- Support for multiple languages.

15. Development Milestones

- **Milestone 1 (Week 4):** Core User Authentication & Profile Module Completion.
- **Milestone 2 (Week 8):** Neighborhood Feed & Basic Posting Functionality.
- **Milestone 3 (Week 12):** MVP Core Features - Chat & Notifications Integration.
- **Milestone 4 (Week 16):** MVP Testing & Quality Assurance.
- **Milestone 5 (Week 18):** MVP Deployment & Pilot Launch.

- **Milestone 6 (Post-MVP):** Development of Phase 2 features (Business Listings, Events, etc.) based on roadmap priorities.

16. Frontend Information Architecture

Public Website Navigation

- **Home**
- **About Us**
- **Features**
- **Pricing**
- **Communities**
- **Businesses**
- **Services**
- **Events**
- **Blog**
- **Careers**
- **Contact Us**

Authentication Module

- **Login**
- **Registration**
- **Forgot Password**
- **OTP Verification**

Mobile App Navigation

- **Home**
- **Discover**
- **Create**

- **Notifications**

- **Profile**

Home Feed

- **Neighborhood Feed**

- **Society Feed**

- **Trending Feed**

- **Nearby Feed**

Discover Module

- **Nearby People**

- **Nearby Businesses**

- **Nearby Events**

- **Nearby Services**

- **Communities**

- **Interest Groups**

Communities Module

- **My Communities**

- **Suggested Communities**

- **Create Community**

Society Module

- **Announcements**

- **Complaints**

- **Parking**

- **Visitors**
- **Documents**
- **Polls**
- **Events**

Business Dashboard

- **Manage Listings**
- **Offers**
- **Analytics**
- **Leads**
- **Reviews**

Admin Panel

- **Dashboard**
- **User Management**
- **Content Moderation**
- **Business Verification**
- **Analytics**

Suggested Flutter Folder Structure

- **core/**
- **features/**
- **shared/**
- **routes/**
- **localization/**

Conclusion

17. Admin Panel Menu Structure

Dashboard

- Overview
- Analytics
- Recent Activities
- Quick Actions

User Management

- All Users
- Residents
- Business Owners
- Service Providers
- Community Leaders
- User Verification
- Blocked Users

Community Management

- Communities List
- Community Requests
- Community Categories

Society Management

- Societies
- Complaints
- Visitors
- Parking
- Polls

Business Management

- Business Listings
- Business Verification
- Categories
- Featured Businesses
- Reviews Management

Service Management

- Service Providers
- Service Categories
- Service Verification
- Ratings & Reviews

Event Management

- All Events
- Create Event
- Event Categories
- RSVP Management

Content Moderation

- Posts Moderation
- Comments Moderation
- Reported Content
- Media Moderation

Advertisement Management

- Banner Ads
- Sponsored Posts
- Campaign Management

Notification Management

- Push Notifications
- Email Notifications
- Broadcast Messages

Reports & Analytics

- User Analytics
- Engagement Reports
- Revenue Reports

CMS Management

- Pages

- Blog Management
- FAQ Management
- Media Library

Settings

- General Settings
- Roles & Permissions
- Payment Settings
- API Configuration

Support & Tickets

- Support Tickets
- Feedback Management

System Logs

- Activity Logs
- Error Logs

This Product Requirements Document provides a comprehensive overview of the proposed hyper-local social media application for India. By focusing on community building and local economic empowerment, the app has the potential to become an indispensable tool for residents and businesses alike. The outlined MVP scope and phased roadmap ensure a structured approach to development and market entry, allowing for iterative improvements based on real-world user feedback.

18. API Endpoint List

Base URL

Production

<https://api.yourapp.com/api/v1>

Staging

<https://staging-api.yourapp.com/api/v1>

18.1 Authentication APIs

Method	Endpoint	Description
POST	/auth/register	Register new user
POST	/auth/login	User login
POST	/auth/logout	Logout user
POST	/auth/send-otp	Send OTP
POST	/auth/verify-otp	Verify OTP
POST	/auth/refresh-token	Refresh JWT Token
POST	/auth/forgot-password	Forgot Password
POST	/auth/reset-password	Reset Password
GET	/auth/me	Logged-in User Profile

18.2 User APIs

Method	Endpoint
GET	/users
GET	/users/{id}
PUT	/users/{id}
DELETE	/users/{id}
PUT	/users/profile
POST	/users/profile-image
GET	/users/nearby
GET	/users/search
POST	/users/follow
DELETE	/users/unfollow/{id}
GET	/users/followers
GET	/users/following

18.3 Community APIs

Method	Endpoint
GET	/communities
POST	/communities
GET	/communities/{id}
PUT	/communities/{id}
DELETE	/communities/{id}
POST	/communities/{id}/join
POST	/communities/{id}/leave
GET	/communities/{id}/members
GET	/communities/my
GET	/communities/suggested
POST	/communities/{id}/invite

18.4 Society APIs

Method	Endpoint
GET	/societies
POST	/societies
GET	/societies/{id}
PUT	/societies/{id}
DELETE	/societies/{id}
GET	/societies/{id}/members
POST	/societies/{id}/join
POST	/societies/{id}/leave
GET	/societies/{id}/announcements
POST	/societies/{id}/announcements
GET	/societies/{id}/complaints
POST	/societies/{id}/complaints
PUT	/societies/complaints/{id}

18.5 Feed/Post APIs

Method	Endpoint
GET	/posts
POST	/posts
GET	/posts/{id}
PUT	/posts/{id}
DELETE	/posts/{id}
POST	/posts/{id}/like
DELETE	/posts/{id}/unlike
POST	/posts/{id}/comment
GET	/posts/{id}/comments
POST	/posts/{id}/share
POST	/posts/{id}/save
DELETE	/posts/{id}/unsave
GET	/posts/trending
GET	/posts/nearby

18.6 Event APIs

Method	Endpoint
GET	/events
POST	/events
GET	/events/{id}
PUT	/events/{id}
DELETE	/events/{id}
POST	/events/{id}/join
POST	/events/{id}/leave
GET	/events/{id}/participants
GET	/events/upcoming
GET	/events/nearby
GET	/events/my

18.7 Business APIs

Method	Endpoint
GET	/businesses
POST	/businesses
GET	/businesses/{id}
PUT	/businesses/{id}
DELETE	/businesses/{id}
GET	/businesses/nearby
GET	/businesses/featured
POST	/businesses/{id}/review
GET	/businesses/{id}/reviews
POST	/businesses/{id}/offer
GET	/businesses/categories

18.8 Service Provider APIs

Method	Endpoint
GET	/services
POST	/services
GET	/services/{id}
PUT	/services/{id}
DELETE	/services/{id}
GET	/services/nearby
POST	/services/{id}/book
GET	/services/bookings
PUT	/services/bookings/{id}
POST	/services/{id}/review

18.9 Chat APIs

Method	Endpoint
GET	/chats
POST	/chats
GET	/chats/{id}
POST	/chats/{id}/message
GET	/chats/{id}/messages
PUT	/messages/{id}/read
DELETE	/messages/{id}

18.10 Notification APIs

Method	Endpoint
GET	/notifications
PUT	/notifications/read-all
PUT	/notifications/{id}/read
DELETE	/notifications/{id}
POST	/notifications/preferences

18.11 Search APIs

Method	Endpoint
GET	/search/global
GET	/search/users
GET	/search/posts
GET	/search/businesses
GET	/search/services
GET	/search/events
GET	/search/communities

18.12 File Upload APIs

Method	Endpoint
POST	/upload/image
POST	/upload/video
POST	/upload/document
DELETE	/upload/{id}

18.13 Report & Moderation APIs

Method	Endpoint
POST	/reports
GET	/reports
PUT	/reports/{id}
DELETE	/reports/{id}

18.14 Admin APIs

Dashboard

Method	Endpoint
GET	/admin/dashboard
GET	/admin/analytics

User Management

Method	Endpoint
GET	/admin/users
PUT	/admin/users/{id}/block
PUT	/admin/users/{id}/unblock
DELETE	/admin/users/{id}

Business Verification

Method	Endpoint
GET	/admin/businesses/pending
PUT	/admin/businesses/{id}/approve
PUT	/admin/businesses/{id}/reject

Content Moderation

Method	Endpoint
GET	/admin/reported-posts
DELETE	/admin/posts/{id}
DELETE	/admin/comments/{id}

CMS APIs

Method	Endpoint
GET	/admin/pages
POST	/admin/pages
PUT	/admin/pages/{id}
DELETE	/admin/pages/{id}

18.15 WebSocket Endpoints

- /ws/chat
- /ws/notifications
- /ws/community-feed
- /ws/event-updates
- /ws/online-users

Standard Response Format

Success Response

```
{  
  "success": true,  
  "message": "Data fetched successfully",  
  "data": {},  
  "errors": null  
}
```

Error Response

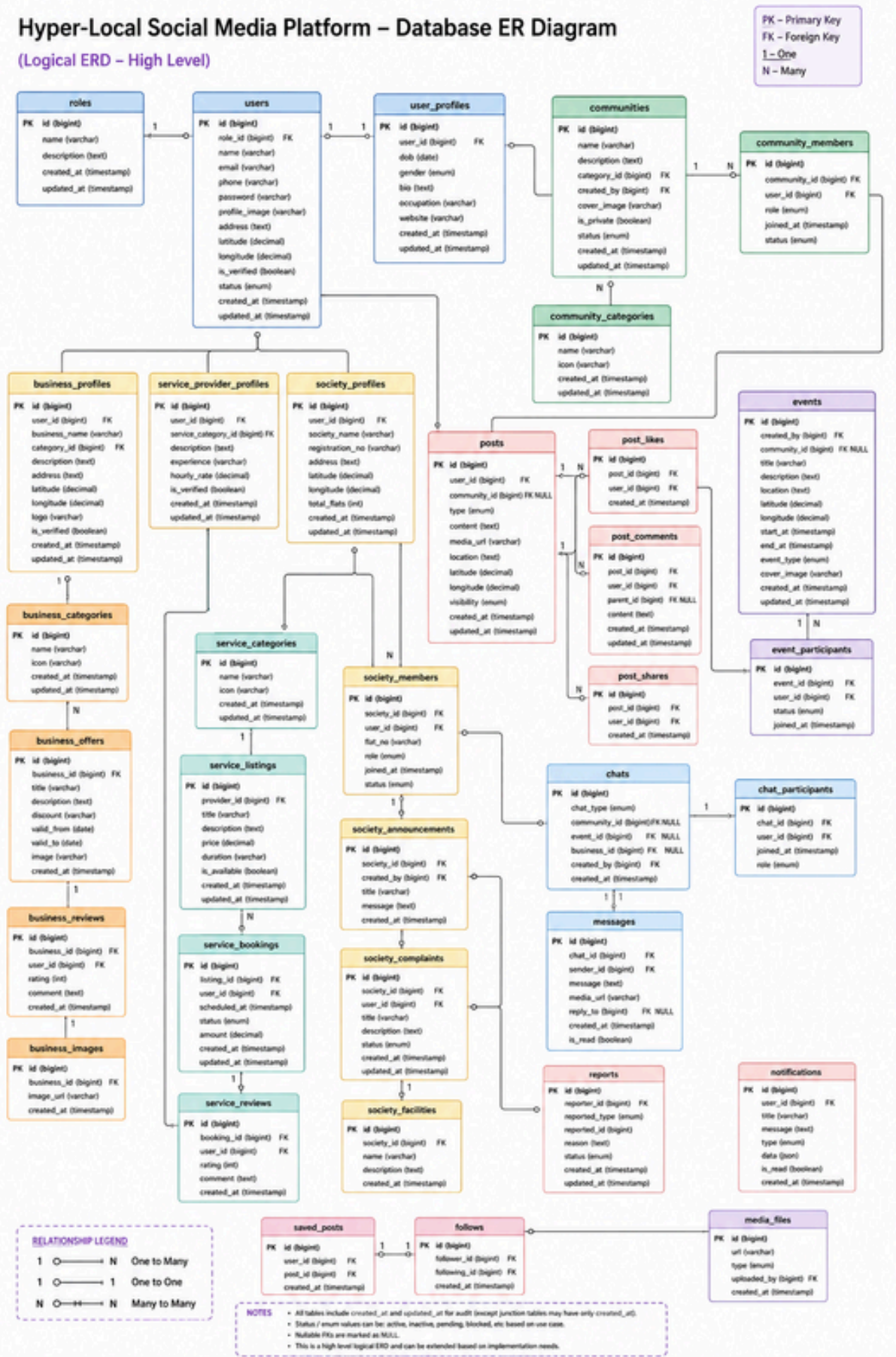
```
{  
  "success": false,  
  "message": "Validation failed",  
  "errors": [  

```

```
{  
  "field": "email",  
  "message": "Email already exists"  
}  
]  
}
```

Hyper-Local Social Media Platform – Database ER Diagram

(Logical ERD – High Level)



19. Role & Permission Matrix

User Roles

Role ID	Role Name	Description
R001	Super Admin	Complete platform access
R002	Admin	Platform operational management
R003	Moderator	Community and content moderation
R004	Resident/User	Standard application user
R005	Community Admin	Community owner/manager
R006	Society Admin	Apartment/Society administrator
R007	Business Owner	Local business owner
R008	Service Provider	Local service provider
R009	Event Organizer	Event creator and manager
R010	Guest User	Non-registered visitor

19.1 Authentication Permissions

Permission	Super Admin	Admin	Moderator	Resident	Community Admin	Business	Service Provider	Guest
Register	✓	✓	✓	✓	✓	✓	✓	✓
Login	✓	✓	✓	✓	✓	✓	✓	✗
Logout	✓	✓	✓	✓	✓	✓	✓	✗
Reset Password	✓	✓	✓	✓	✓	✓	✓	✗
Manage Own Profile	✓	✓	✓	✓	✓	✓	✓	✗

19.2 Feed & Posts Permissions

Permission	SA	Admin	Mod	Resident	Community Admin	Business	Service
Create Post	✓	✓	✓	✓	✓	✓	✓
Edit Own Post	✓	✓	✓	✓	✓	✓	✓
Delete Own Post	✓	✓	✓	✓	✓	✓	✓
Delete Any Post	✓	✓	✓	✗	✗	✗	✗
Like Posts	✓	✓	✓	✓	✓	✓	✓
Comment	✓	✓	✓	✓	✓	✓	✓
Share Posts	✓	✓	✓	✓	✓	✓	✓
Pin Post	✓	✓	✓	✗	✓	✗	✗

19.3 Community Permissions

Permission	SA	Admin	Mod	Resident	Community Admin
Create Community	✓	✓	✓	✓	✓
Update Community	✓	✓	✓	✗	✓
Delete Community	✓	✓	✗	✗	✓
Approve Members	✓	✓	✓	✗	✓
Remove Members	✓	✓	✓	✗	✓
Create Polls	✓	✓	✓	✓	✓
Create Announcement	✓	✓	✓	✗	✓

19.4 Society Permissions

Permission	SA	Admin	Society Admin	Resident
Create Society	✓	✓	✓	✗
Edit Society	✓	✓	✓	✗
Add Members	✓	✓	✓	✗
Manage Parking	✓	✓	✓	✗
Create Complaint	✓	✓	✓	✓
Resolve Complaint	✓	✓	✓	✗
Visitor Management	✓	✓	✓	✗
Publish Notices	✓	✓	✓	✗

19.5 Business Permissions

Permission	SA	Admin	Business Owner
Create Business Profile	✓	✓	✓
Edit Own Business	✓	✓	✓
Delete Own Business	✓	✓	✓
Verify Business	✓	✓	✗
Create Offers	✓	✓	✓
View Analytics	✓	✓	✓
View Leads	✓	✓	✓
Respond Reviews	✓	✓	✓

19.6 Service Provider Permissions

Permission	SA	Admin	Service Provider
Create Service Listing	✓	✓	✓
Edit Listing	✓	✓	✓
Delete Listing	✓	✓	✓
Manage Availability	✓	✓	✓
Accept Booking	✓	✓	✓
Reject Booking	✓	✓	✓
View Earnings	✓	✓	✓

19.7 Event Permissions

Permission	SA	Admin	Organizer	Resident
Create Event	✓	✓	✓	✓
Edit Event	✓	✓	✓	✗
Delete Event	✓	✓	✓	✗
Cancel Event	✓	✓	✓	✗
View Participants	✓	✓	✓	✗
RSVP Event	✓	✓	✓	✓

19.8 Chat Permissions

Permission	All Registered Users
One-to-One Chat	✓
Group Chat	✓
Send Media	✓
Delete Own Message	✓
Report Chat	✓

19.9 Moderation Permissions

Permission	SA	Admin	Moderator
Review Reported Content	✓	✓	✓
Remove Content	✓	✓	✓
Suspend User	✓	✓	✗
Block User	✓	✓	✗
Ban Community	✓	✓	✗
Review Appeals	✓	✓	✓

19.10 CMS Permissions

Permission	SA	Admin
Create Pages	✓	✓
Edit Pages	✓	✓
Delete Pages	✓	✓
Manage FAQ	✓	✓
Manage Banners	✓	✓
Manage Blogs	✓	✓

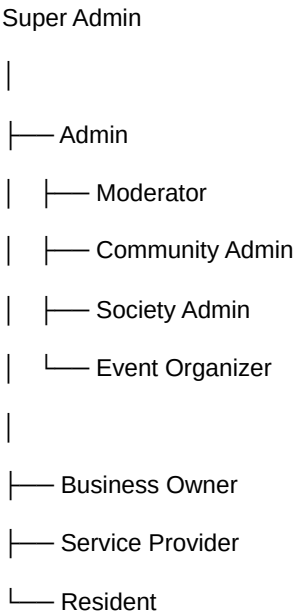
19.11 System Administration Permissions

Permission	Super Admin
Manage Admin Users	✓
Create Roles	✓
Assign Permissions	✓
Configure APIs	✓
Configure Payment Gateway	✓
View Audit Logs	✓
View System Logs	✓
Backup & Restore	✓
Configure Platform Settings	✓

19.12 Permission Legend

Symbol	Meaning
✓	Allowed
✗	Not Allowed
SA	Super Admin

Recommended RBAC Architecture



Authorization Strategy

- Use **Role-Based Access Control (RBAC)**.
- Support **Permission-Based Access Control (PBAC)** for advanced modules.
- Implement JWT Claims-based authorization.
- Store permissions in database tables:
 - Roles
 - Permissions
 - RolePermissions
 - UserRoles
 - UserPermissions

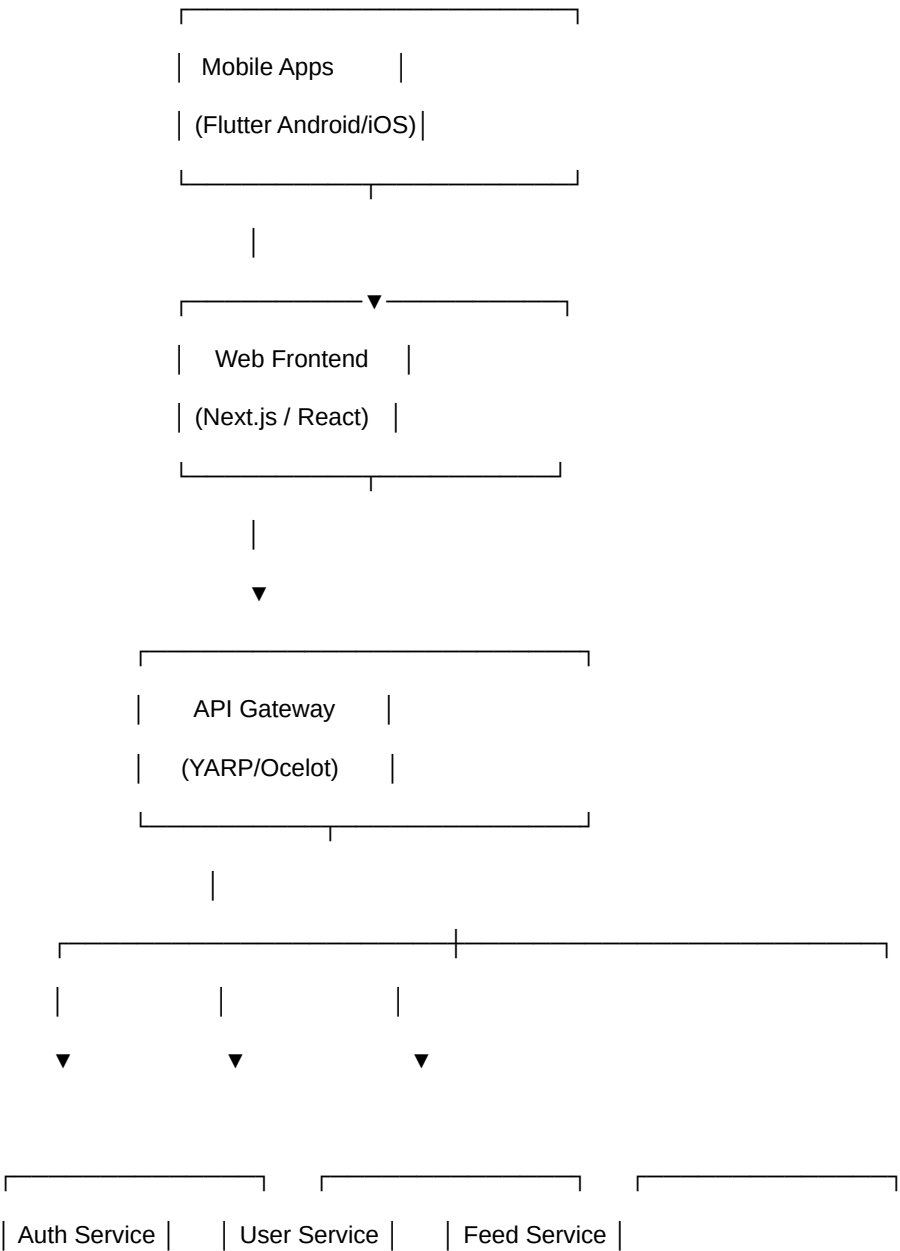
This matrix should be implemented both in **Backend APIs** and **Frontend UI Guards**.

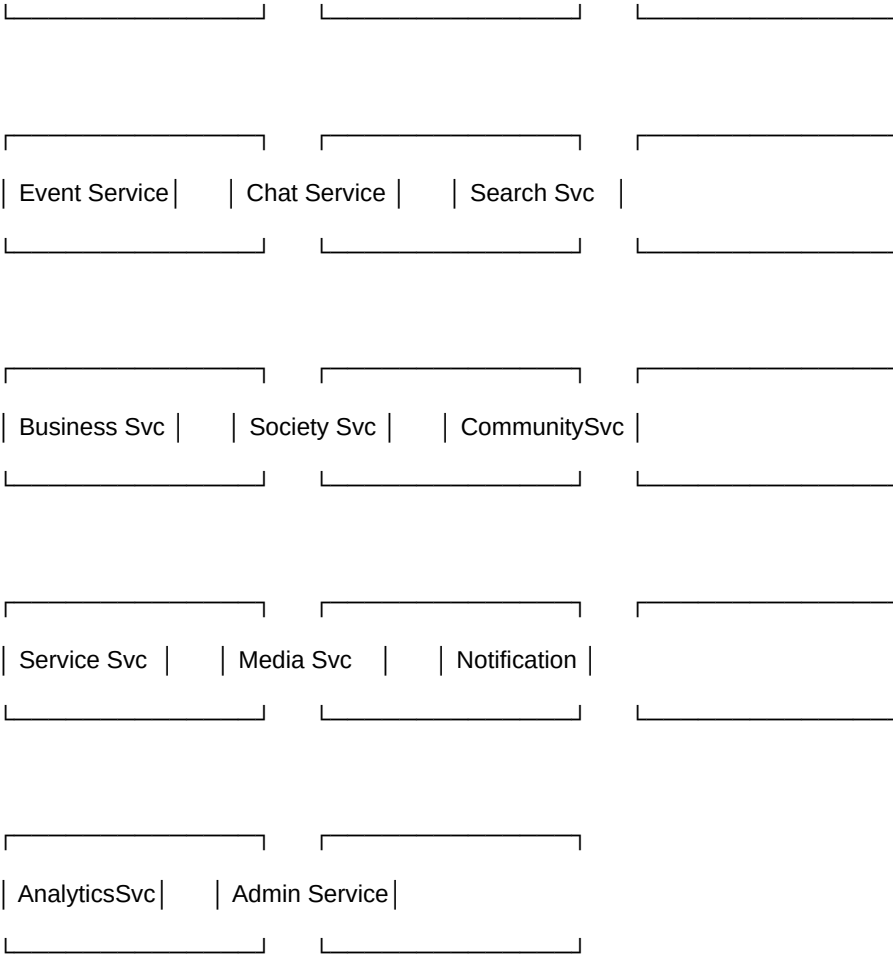
20. Microservices Architecture

20.1 Architecture Overview

The Hyper-Local Social Media Platform will follow a **Cloud Native Microservices Architecture** using **ASP.NET Core**, **PostgreSQL**, **Redis**, **RabbitMQ/Kafka**, **Docker**, and **Kubernetes**.

High-Level Architecture





20.2 Infrastructure Layer

Client Applications

Mobile Applications

- Flutter Android App
- Flutter iOS App

Web Applications

- React.js User Portal
- React.js Admin Portal

20.3 API Gateway

Recommended

Option 1

- YARP Reverse Proxy (.NET)

Option 2

- Ocelot API Gateway

Responsibilities

- Authentication Validation
- JWT Verification
- Request Routing
- Rate Limiting

- API Aggregation
- Load Balancing
- CORS Handling
- API Versioning

20.4 Core Microservices

1. Authentication Service

Responsibilities

- Registration
- Login
- OTP Verification
- JWT Generation
- Refresh Tokens
- Password Reset
- Social Login

Database

PostgreSQL

Redis Cache

2. User Service

Responsibilities

- User Profile
- User Preferences
- Followers
- Following
- Interests
- Geo Location

Tables

- Users
- UserProfiles
- UserSettings
- UserInterests

3. Community Service

Responsibilities

- Community Creation
- Join Requests
- Member Management
- Community Roles
- Community Feed Mapping

4. Society Service

Responsibilities

- Society Management
- Announcements
- Complaints
- Parking
- Visitors

- Staff Management

5. Feed Service

Responsibilities

- Posts
- Likes
- Comments
- Shares
- Saved Posts
- Trending Feed

Cache

Redis

6. Event Service

Responsibilities

- Event Management
- RSVP
- Event Invitations
- Event Reminders

7. Business Service

Responsibilities

- Business Listings
- Offers
- Reviews
- Promotions

8. Service Marketplace Service

Responsibilities

- Service Listings
- Bookings
- Availability
- Payments

9. Chat Service

Technology

- SignalR
- WebSocket

Responsibilities

- Private Chat
- Group Chat
- Message Delivery
- Read Receipts

10. Notification Service

Channels

- Push Notification

- SMS
- Email
- In-App Notification

Providers

- Firebase FCM
- Twilio
- SendGrid

11. Search Service

Technology

Elasticsearch

Responsibilities

- Global Search
- Nearby Search
- Autocomplete
- Geo Search

12. Media Service

Responsibilities

- Image Upload
- Video Upload
- Compression
- CDN Delivery

Storage

AWS S3 + CloudFront

13. Analytics Service

Responsibilities

- User Analytics
- Engagement Analytics
- Revenue Analytics
- Community Analytics

14. Admin Service

Responsibilities

- Admin Dashboard
- Moderation
- CMS
- Reports
- Configuration

20.5 Event Driven Architecture

Message Broker

Recommended

RabbitMQ

or

Apache Kafka

Sample Events

- UserRegisteredEvent
- PostCreatedEvent
- CommentAddedEvent
- EventCreatedEvent
- BookingCompletedEvent
- BusinessVerifiedEvent
- NotificationTriggeredEvent

20.6 Databases Per Service

Service	Database
Auth Service	PostgreSQL
User Service	PostgreSQL
Feed Service	PostgreSQL
Chat Service	MongoDB
Notification Service	PostgreSQL
Search Service	Elasticsearch
Analytics Service	ClickHouse
Media Service	AWS S3

20.7 Caching Layer

Redis Usage

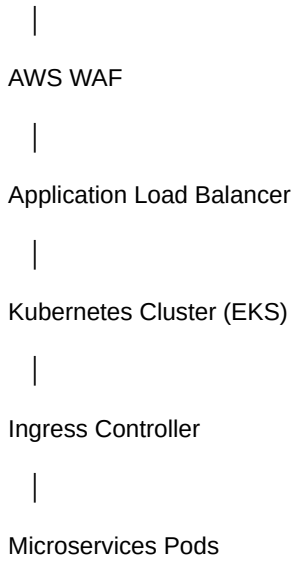
- User Session Cache
- Feed Cache
- Trending Posts Cache
- Search Cache
- OTP Cache
- Rate Limiting Cache

20.8 Kubernetes Deployment Architecture

Internet

|

CloudFront CDN



20.9 Observability Stack

Monitoring

- Prometheus
- Grafana

Logging

- ELK Stack
 - Elasticsearch
 - Logstash
 - Kibana

Tracing

- OpenTelemetry
- Jaeger

20.10 Security Architecture

- OAuth2
- JWT Authentication
- HTTPS Everywhere
- API Gateway Security
- Rate Limiting
- WAF
- Encryption at Rest
- Encryption in Transit
- Role Based Access Control

20.11 Recommended Technology Stack

Layer	Technology
Mobile	Flutter
Frontend	React.js / Next.js
Backend	ASP.NET Core
API Gateway	YARP/Ocelot
Database	PostgreSQL
Cache	Redis
Search	Elasticsearch
Messaging	RabbitMQ
Realtime	SignalR
Storage	AWS S3
Container	Docker
Orchestration	Kubernetes
Monitoring	Prometheus + Grafana

20.12 MVP Microservices

For MVP launch, start with:

1. API Gateway
2. Auth Service
3. User Service
4. Feed Service
5. Community Service
6. Chat Service
7. Notification Service
8. Media Service

Additional services can be introduced gradually as the platform scales.

21. Screen-wise User Flow

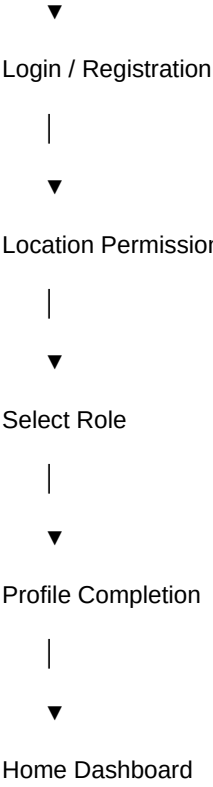
21.1 Overall Application Flow

Splash Screen



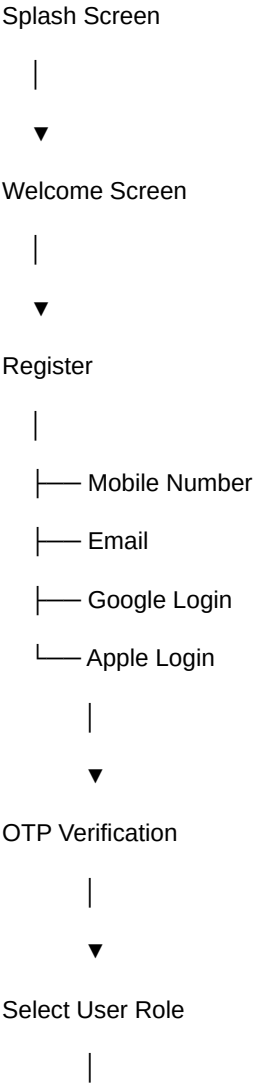
Onboarding Screens

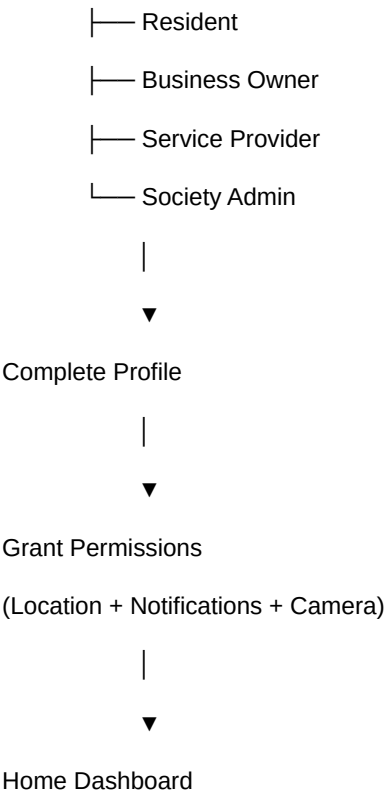




21.2 Authentication Flow

New User Flow



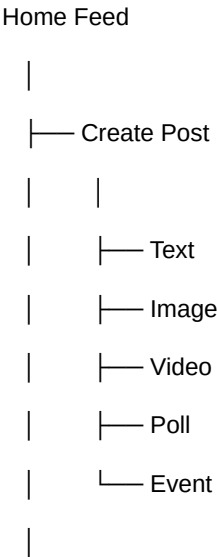


21.3 Resident User Flow

Home Navigation



Feed Flow



- └─ Like Post
- └─ Comment
- └─ Share
- └─ Save
- └─ Report

21.4 Community User Flow

Communities

- |
- └─ Suggested Communities
- └─ My Communities
- └─ Search Communities
- └─ Create Community
- |
- ▼

Community Details

- |
- └─ Feed
- └─ Members
- └─ Events
- └─ Files
- └─ Polls
- └─ Gallery
- └─ Chat

21.5 Society Management Flow

Society Dashboard

- |
- └─ Announcements
- └─ Complaints
- | |
- | └─ Create Complaint
- | └─ View Status
- | └─ Close Complaint
- |
- └─ Parking
- └─ Visitors

- └─ Polls
- └─ Documents
- └─ Emergency Contacts

21.6 Event Module Flow

Events

- |
- └─ Nearby Events
- └─ Upcoming Events
- └─ My Events
- └─ Create Event
 - |
 - ▼

Event Details

- |
- └─ RSVP
- └─ Interested
- └─ Share
- └─ Invite Friends
- └─ Event Chat

21.7 Business Owner Flow

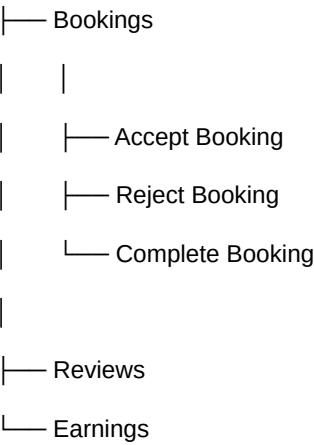
Business Dashboard

- |
- └─ Create Business Profile
- └─ Add Products
- └─ Add Offers
- └─ Manage Reviews
- └─ Analytics
- └─ Leads
- └─ Subscription

21.8 Service Provider Flow

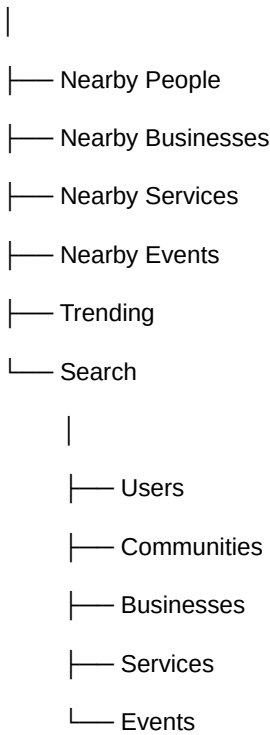
Service Dashboard

- |
- └─ Create Service
- └─ Manage Availability



21.9 Discover Flow

Discover Screen



21.10 Chat Flow

Chat List



Chat Window



- └─ Text Message
- └─ Image Message
- └─ Video Message
- └─ Voice Message
- └─ File Sharing
- └─ Report User

21.11 Notification Flow

Notifications

- |
- └─ Likes
- └─ Comments
- └─ Follow Requests
- └─ Event Reminders
- └─ Community Updates
- └─ Chat Messages
- └─ Promotions

21.12 Profile Flow

Profile

- |
- └─ Edit Profile
- └─ My Posts
- └─ Saved Posts
- └─ Followers
- └─ Following
- └─ My Communities
- └─ My Events
- └─ Settings
- └─ Logout

21.13 Admin Panel Flow

Admin Login

- |
- ▼

Dashboard

- |

- |— User Management
- |— Community Management
- |— Society Management
- |— Business Management
- |— Service Management
- |— Event Management
- |— Content Moderation
- |— Advertisement Management
- |— Notifications
- |— Reports
- |— CMS
- |— Analytics
- |— Settings
- |— Audit Logs

21.14 Super Admin Flow

Super Admin Dashboard

- |
- |— Manage Admins
- |— Roles & Permissions
- |— Platform Configuration
- |— API Configuration
- |— Subscription Plans
- |— Payment Gateway
- |— Backup & Restore
- |— System Monitoring

21.15 Guest User Flow

Landing Page

- |
- |— Browse Public Communities
- |— Browse Events
- |— Browse Businesses
- |— Register/Login

21.16 Key Screens List

Authentication (10 Screens)

1. Splash Screen
2. Onboarding
3. Login
4. Register
5. OTP Verification
6. Forgot Password
7. Reset Password
8. Select Role
9. Complete Profile
10. Permissions Screen

Main App (50+ Screens)

- Home Feed
- Discover
- Community List
- Community Details
- Society Dashboard
- Events
- Chat
- Notifications
- Profile
- Settings
- Business Dashboard
- Service Dashboard

Admin Portal (30+ Screens)

- Dashboard
- User Management
- Content Moderation
- Analytics
- CMS
- Reports
- Settings

Estimated Total Screens: 100–120 Screens

22. Notification Matrix

22.1 Notification Channels

Channel	Description
In-App Notification	Notification displayed inside app
Push Notification	Mobile push notification
Email Notification	Notification sent via email
SMS Notification	SMS notification
Real-Time Notification	Instant notification via WebSocket/SignalR

22.2 User Authentication Notifications

Event	In-App	Push	Email	SMS	Real-Time
User Registration Success	✓	✓	✓	✓	✗
OTP Sent	✗	✗	✓	✓	✗
OTP Verified	✓	✗	✗	✗	✗
Login From New Device	✓	✓	✓	✗	✗
Password Reset	✓	✓	✓	✓	✗
Password Changed	✓	✓	✓	✗	✗

22.3 Social & Feed Notifications

Event	In-App	Push	Email	SMS	Real-Time
Someone Liked My Post	✓	✓	✗	✗	✓
Someone Commented on My Post	✓	✓	✗	✗	✓
Someone Shared My Post	✓	✓	✗	✗	✓
Mentioned in Post	✓	✓	✗	✗	✓
New Follower	✓	✓	✗	✗	✓
Follow Request	✓	✓	✗	✗	✓

22.4 Community Notifications

Event	In-App	Push	Email	SMS	Real-Time
Community Join Request	✓	✓	✗	✗	✓
Community Join Approved	✓	✓	✗	✗	✓
New Community Post	✓	Optional	✗	✗	✓
New Poll Created	✓	✓	✗	✗	✓
New Announcement	✓	✓	✓	Optional	✓
Removed from Community	✓	✓	✗	✗	✗

22.5 Society Notifications

Event	In-App	Push	Email	SMS	Real-Time
New Society Notice	✓	✓	✓	Optional	✓
Complaint Status Updated	✓	✓	✓	✗	✓
Visitor Arrival	✓	✓	✗	Optional	✓
Parking Allocation Updated	✓	✓	✗	✗	✓
Maintenance Due Reminder	✓	✓	✓	✓	✗
Emergency Alert	✓	✓	✓	✓	✓

22.6 Event Notifications

Event	In-App	Push	Email	SMS	Real-Time
Event Created	✓	✓	✗	✗	✓
Event Reminder	✓	✓	✓	Optional	✗
RSVP Confirmed	✓	✓	✓	✗	✗
Event Cancelled	✓	✓	✓	✓	✓
Event Updated	✓	✓	✓	✗	✓
Invited to Event	✓	✓	✗	✗	✓

22.7 Chat Notifications

Event	In-App	Push	Email	SMS	Real-Time
New Private Message	✓	✓	✗	✗	✓
New Group Message	✓	✓	✗	✗	✓
Voice Call	✓	✓	✗	✗	✓
Video Call	✓	✓	✗	✗	✓
Message Read	✓	✗	✗	✗	✓

22.8 Business Notifications

Event	In-App	Push	Email	SMS	Real-Time
New Review Received	✓	✓	✓	✗	✓
New Lead Generated	✓	✓	✓	Optional	✓
Offer Expiry Reminder	✓	✓	✓	✗	✗
Business Verification Approved	✓	✓	✓	✓	✗
Business Verification Rejected	✓	✓	✓	✗	✗

22.9 Service Provider Notifications

Event	In-App	Push	Email	SMS	Real-Time
New Booking Request	✓	✓	✓	Optional	✓
Booking Accepted	✓	✓	✓	✗	✓
Booking Rejected	✓	✓	✓	✗	✓
Booking Completed	✓	✓	✓	✗	✓
Payment Received	✓	✓	✓	✓	✗

22.10 Moderation Notifications

Event	In-App	Push	Email	SMS	Real-Time
Content Reported	✓	✓	✗	✗	✓
Content Removed	✓	✓	✓	✗	✗
User Account Suspended	✓	✓	✓	✓	✗
Community Suspended	✓	✓	✓	✗	✗

22.11 Admin Notifications

Event	In-App	Push	Email	SMS	Real-Time
New User Registration	✓	✗	Optional	✗	✓
New Business Verification Request	✓	✓	✓	✗	✓
New Report Received	✓	✓	✗	✗	✓
System Error Alert	✓	✓	✓	✓	✓
Payment Failure	✓	✓	✓	Optional	✓

22.12 Notification Priority Matrix

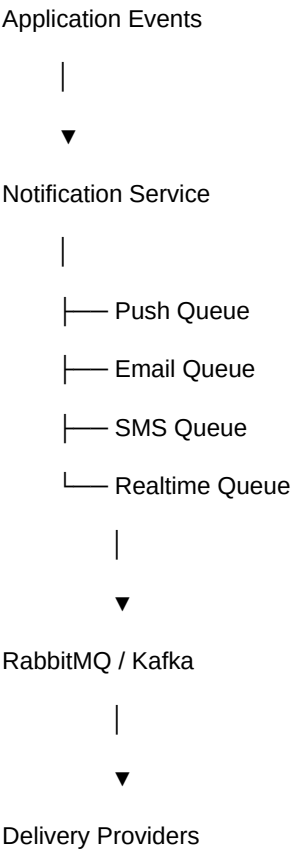
Priority	Type	Examples
Critical	Immediate	Emergency Alerts, System Failure
High	Within Seconds	Chat, Calls, Event Cancellation
Medium	Within Minutes	Comments, Likes, Community Updates
Low	Scheduled	Promotions, Newsletters

22.13 User Notification Preferences

Users can configure:

- Enable/Disable Push Notifications
- Enable/Disable Email Notifications
- Enable/Disable SMS Notifications
- Mute Communities
- Mute Users
- Quiet Hours (Do Not Disturb)
- Notification Frequency
 - Instant
 - Hourly Digest
 - Daily Digest
 - Weekly Digest

22.14 Backend Architecture for Notifications



22.15 Notification Providers

Channel	Provider
Push	Firebase Cloud Messaging (FCM)
Email	SendGrid / AWS SES
SMS	Twilio / MSG91
Real-Time	SignalR / WebSocket

22.16 Database Tables

- Notifications
- NotificationTemplates
- NotificationPreferences
- NotificationLogs
- PushTokens
- EmailTemplates
- SmsTemplates

This notification matrix should be implemented in both the **Notification Service** and **User Preference Module**.

23. Sprint Planning (Phase-wise Development)

Project Duration

Estimated MVP Duration: 24 Weeks (6 Months)

Sprint Duration: 2 Weeks

Total Sprints (MVP): 12 Sprints

Team Composition

Role	Count
Product Manager	1
Business Analyst	1
UI/UX Designer	2
Flutter Developers	3
Backend Developers (.NET)	3
QA Engineers	2
DevOps Engineer	1
AI/ML Engineer	1
Project Manager	1

Phase 0: Discovery & Planning

Duration

Week 1–2

Sprint 0

Activities

- Requirement Gathering
- Competitor Analysis
- User Persona Creation
- Product Roadmap
- Architecture Design
- Technology Finalization
- PRD Signoff

Deliverables

- BRD

- PRD
- SRS
- Wireframes
- Architecture Document

Phase 1: UI/UX Design

Duration

Week 3–6

Sprint 1

Activities

- Information Architecture
- User Flow Design
- Low Fidelity Wireframes

Screens

- Authentication
- Home
- Feed
- Discover
- Profile

Deliverables

- Wireframes

Sprint 2

Activities

- High Fidelity UI Design
- Design System
- Component Library

Screens

- Community Module
- Event Module
- Chat Module
- Business Module

Deliverables

- Figma Design
- UI Kit

Phase 2: Foundation Setup

Duration

Week 7–8

Sprint 3

Backend

- Repository Setup

- Microservices Setup
- API Gateway Setup
- Database Setup

Frontend

- Flutter Project Setup
- State Management Setup
- Navigation Architecture

DevOps

- CI/CD Setup
- Docker Setup

Deliverables

- Development Environment Ready

Phase 3: Authentication & User Management

Duration

Week 9–10

Sprint 4

Backend

- Authentication APIs
- JWT
- OTP Login
- User APIs

Frontend

- Splash Screen
- Login
- Registration
- Profile Completion

QA

- Authentication Testing

Deliverables

- Authentication Module Complete

Phase 4: Feed & Social Features

Duration

Week 11–14

Sprint 5

Features

- Home Feed
- Create Post
- Like/Comment/Share
- Media Upload

Deliverables

- Feed Module Complete

Sprint 6

Features

- Followers
- Following
- User Search
- Nearby Users

Deliverables

- Social Networking Features

Phase 5: Communities & Society

Duration

Week 15–18

Sprint 7

Features

- Community Creation
- Community Feed
- Community Chat

Deliverables

- Community Module

Sprint 8

Features

- Society Dashboard
- Complaints
- Announcements
- Polls
- Visitors

Deliverables

- Society Module

Phase 6: Events & Chat

Duration

Week 19–20

Sprint 9

Features

- Event Creation
- RSVP
- Event Invitations

Features

- Real-Time Chat
- Private Chat
- Group Chat

Deliverables

- Event + Chat Module

Phase 7: Business & Services Marketplace

Duration

Week 21–22

Sprint 10

Features

- Business Listing
- Offers
- Reviews

Features

- Service Listings
- Booking System

Deliverables

- Marketplace Module

Phase 8: Notifications & Admin Panel

Duration

Week 23–24

Sprint 11

Features

- Push Notifications
- Email Notifications
- In-App Notifications

Features

- Admin Dashboard
- CMS
- Moderation

Deliverables

- Admin Portal

Phase 9: Testing, Security & Release

Duration

Week 25–26

Sprint 12

Activities

- Functional Testing
- Regression Testing
- Security Testing
- Performance Testing
- UAT

Deliverables

- Production Release Candidate

MVP Scope

Included

User Features

- ✓ Authentication
- ✓ Profile Management
- ✓ Feed
- ✓ Communities
- ✓ Society Module
- ✓ Events
- ✓ Chat
- ✓ Notifications
- ✓ Business Listings
- ✓ Service Marketplace

Admin Features

- ✓ Dashboard
- ✓ Moderation
- ✓ Analytics
- ✓ CMS

Post MVP Roadmap (Phase 2)

Month 7–9

AI Features

- AI Recommendations
- AI Moderation
- AI Assistant

Monetization

- Paid Promotions
- Featured Businesses
- Subscription Plans

Month 10–12

Advanced Features

- Video Calling
- Live Streaming
- Digital Payments
- Local Commerce

Agile Ceremony Schedule

Ceremony	Frequency
Sprint Planning	Every 2 Weeks
Daily Standup	Daily
Sprint Review	End of Sprint
Sprint Retrospective	End of Sprint
Backlog Grooming	Weekly

Release Plan

Release	Timeline
Alpha	Week 12
Beta	Week 20
UAT	Week 24
Production MVP	Week 26

Success Metrics

Technical

- API Response < 300 ms
- Crash Free Rate > 99.5%
- Uptime > 99.9%

Business

- 10,000 Users in First 6 Months
- 1,000 Active Communities
- 500 Verified Businesses

Risk Matrix

Risk	Mitigation
Scope Creep	Strict Sprint Planning
Delayed UI	Parallel Design Team
Scalability Issues	Microservices Architecture
Low Adoption	Early Beta Testing

This sprint plan is optimized for **Agile Scrum methodology** and supports a scalable MVP launch within 6 months.